

ASSIGNMENT 5: Multi-Frequency Backward Propagation

Due: Monday, November 10

Objective: The objective of this programming assignment is to experience image formation by multi-frequency backward propagation and to observe the effects on image resolution in the form of *aperture size*.

The system configuration is the same as that in *Assignment 2*, with 4 point sources at the same locations. The *receiver aperture* is a centered linear receiver array with a span of $60\lambda_0$ (from $x = -30\lambda_0$ to $x = +30\lambda_0$), located along $y = -60\lambda_0$, with quarter-wavelength ($\lambda_0/4$) spacing. For each coherent frequency, there are 241 wavefield samples in total over the $60\lambda_0$ -long aperture.

In *Assignment 2*, one single wavelength was applied. As we move up to the multi-frequency (wideband) operating mode, we collect wavefield samples over a wide range of spectrum corresponding to 64 different wavelengths, in the form

$$\lambda_n = 64 \lambda_0 / (n+32) \quad n = 1, 2, \dots, 64$$

Thus, this imaging modality operates with a sequence of wavelengths, from $0.67\lambda_0$ to $2\lambda_0$, corresponding to the spatial-frequency band from $0.5(1/\lambda_0)$ to $1.5(1/\lambda_0)$ resulting a spatial-frequency bandwidth of $(1/\lambda_0)$.

The complete data set is a (64×241) array, corresponding to 64 wavelengths and 241 receiver positions.

Superposition of range profiles:

This is to perform range estimation 241 times at the 241 receiver positions, with the same receiver array configuration. This process produces 241 range profiles over the source region, $\hat{s}_n(x, y)$, for $n = 1, 2, \dots, 241$.

The goal is to perform multi-frequency range estimation of the $(60\lambda_0 \times 60\lambda_0)$ 2D source region with 241 sets of wavefield samples corresponding to the 241 receiver positions. Each range estimate is formed from 64 wavefield samples corresponding to the 64 operating wavelengths.

- (1) By repeating the range estimation procedure, produce 241 range profiles over the source region.
- (2) Superimpose the 241 sub-range profiles *sequentially* and observe the convergence to the overall image,

$$\hat{S}_n(x, y) = \sum_{k=1}^n \hat{s}_k(x, y) \quad n = 1, 2, \dots, 241.$$

- (3) Produce the spectra of the complex range profiles $\hat{S}_n(x, y)$, $n = 1, 2, \dots, 241$
- (4) Compile a 241-frame video of the range-profile sequence $\hat{S}_n(x, y)$.
- (5) Compile a 241-frame video of the spectra of the range-profile sequence.

Report format:

- (A) Cover page.
- (B) Video sequences.
- (C) Summary: (comments based on your observations)
- (D) Appendix: (computer code)